

Identification of Halide Ions (Cl^- , Br^- and I^-)**Aim**

To find out whether chloride, bromide and iodide ions are present in the given salt solutions by carrying out a simple precipitation test.

Requirements

Test tubes, a test tube stand, a dropper, a beaker, etc.

Chemicals Required

Silver nitrate solution (AgNO_3), and separate solutions of potassium chloride, potassium bromide and potassium iodide.

Procedure

1. Take three clean test tubes and mark them as A, B and C.
2. Pour about 5 ml of potassium chloride solution into test tube A, potassium bromide solution into test tube B, and potassium iodide solution into test tube C.
3. Add 4 to 5 drops of silver nitrate solution to each of the three test tubes and mix well.
4. Note down the colour of the precipitate that forms in each test tube.
5. Using the colour of the precipitate obtained, determine which halide ion is present in each solution.

Diagram

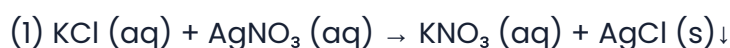
(Label the components in the diagram below)

Diagram will be added here

Observation Table

Test Tube	Reaction / Test	Colour of Precipitate	Inference
A	$\text{KCl} + \text{AgNO}_3$	White	Chloride (Cl^-) present.
B	$\text{KBr} + \text{AgNO}_3$	Light yellow	Bromide (Br^-) present.
C	$\text{KI} + \text{AgNO}_3$	Bright yellow	Iodide (I^-) present.

Chemical Reactions



(Potassium chloride) + (Silver nitrate) $\rightarrow$ (Potassium nitrate) + (White precipitate of silver chloride)



(Potassium bromide) + (Silver nitrate) $\rightarrow$ (Potassium nitrate) + (Light yellow precipitate of silver bromide)



(Potassium iodide) + (Silver nitrate) $\rightarrow$ (Potassium nitrate) + (Bright yellow precipitate of silver iodide)

Inferences

- Halide ions (Cl^- , Br^- , I^-) are precipitated in all the three reactions of above experiments.
- In the above experiment, depending on the colour of the precipitates, presence of Cl^- , Br^- and I^- in the given solutions are confirmed.
- Elements of halogen group in the periodic table show similar properties.

Multiple Choice Questions

1. The number of electrons in the outermost shell of halogens is

(A) 1 (B) 2 (C) 3 (D) 7

Answer: (D) 7

2. is in solid state at room temperature.

(A) Fluorine (B) Iodine (C) Chlorine (D) Bromine

Answer: (B) Iodine

3. Which of the following substances is the strongest reducing agent?

(A) Cl_2 (B) Cl^- ion (C) Br (D) Br^- ion

Answer: (D) Br^- ion

Remember

1. All halogen elements are oxidizing agents.
2. Precipitation reactions produce insoluble salts.

Date: _____

Teacher's Signature: _____